



Problem Based Exercise: NDT-Based Condition Assessment of an Existing Reinforced Concrete Structure

Course Code/Name: 2093116/BMCT

Year/Sem: SE/ III

Date: 12/10/2026

Course Outcome: 2093116.5 Use non-destructive testing methods to analyze the durability and strength of existing concrete structures.

Question				BL									
A 25-year-old reinforced concrete beam-column frame in an institutional building has developed visible cracks, surface deterioration and localized dampness. The building owner plans to continue using the structure for the next 15 years and has requested a preliminary assessment of the concrete quality without damaging the structural members. A team has carried out Rebound Hammer Test and Ultrasonic Pulse Velocity (UPV) Test at selected locations. The following results were obtained:				Analyze									
Location	Rebound Hammer Average	UPV (km/s)	Visual Observation										
Column C1	39	4.25	Good surface condition										
Column C2	31	3.45	Minor cracking										
Beam B1	36	3.90	No visible cracks										
Beam B2	27	2.85	Several cracks and dampness										
Slab S1	34	3.70	Minor surface deterioration										
Slab S2	24	2.50	Cracks and localized spalling										
For preliminary interpretation, assume the following UPV classification:													
<table><tr><th>UPV</th><th>Concrete Quality</th></tr><tr><td>Above 4.0 km/s</td><td>Excellent</td></tr><tr><td>3.5–4.0 km/s</td><td>Good</td></tr><tr><td>3.0–3.5 km/s</td><td>Medium</td></tr><tr><td>Below 3.0 km/s</td><td>Doubtful/Poor</td></tr></table>		UPV	Concrete Quality		Above 4.0 km/s	Excellent	3.5–4.0 km/s	Good	3.0–3.5 km/s	Medium	Below 3.0 km/s	Doubtful/Poor	
UPV	Concrete Quality												
Above 4.0 km/s	Excellent												
3.5–4.0 km/s	Good												
3.0–3.5 km/s	Medium												
Below 3.0 km/s	Doubtful/Poor												
The rebound hammer results indicate that higher rebound numbers generally represent higher surface hardness, but the results must be interpreted along with UPV and site observations. Students shall submit a 2–3-page NDT Assessment Report containing: • Problem description • NDT data interpretation • UPV classification table • Rebound Hammer interpretation • Identification of critical locations • Engineering justification • Recommendation for additional investigations • Final recommendation • Conclusion													



Vidyavardhini's College of Engineering & Technology

Department of Civil Engineering

Academic Year: 2026-27

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Name of Student:			
Roll No. :		Group No.:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
NDT Data Interpretation	2	
Identification of Critical Locations	2	
Engineering Analysis & Justification	2	
Recommendation of Further NDT/Action	2	
Presentation, Neatness & Timely Submission	2	
Total	10	

Performance Indicator	Exceed Expectations (EE) (2 marks)	Meet Expectations (ME) (1 marks)	Below Expectations (BE) (0 marks)
NDT Result Interpretation	Correctly interprets Rebound Hammer and UPV results and accurately assesses the concrete condition.	Interprets most NDT results correctly with minor errors.	Interpretation is incorrect or incomplete.
Concrete Condition Assessment	Correctly identifies critical locations and evaluates concrete strength and durability using combined NDT and visual observations.	Identifies major problem locations with reasonable justification.	Unable to correctly identify or justify critical locations.
Engineering Decision & Justification	Provides a clear and technically sound decision supported by relevant NDT findings.	Provides an appropriate decision with adequate justification.	Decision is unclear, inappropriate, or poorly justified.
Recommendation for Further Investigation	Recommend suitable additional NDT/assessment methods and explain their purpose clearly.	Recommends generally suitable methods with limited explanation.	Recommendations are unsuitable, unclear, or missing.
Report Quality, Neatness & Timely Submission	Report is well-organized, neat, technically prese	----	---

Checked by

Name of Faculty : Mr. Arbaz M. Kazi

Signature :